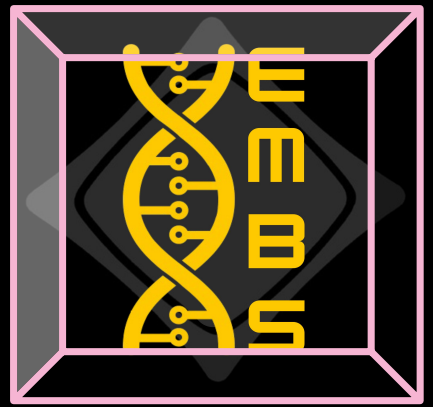
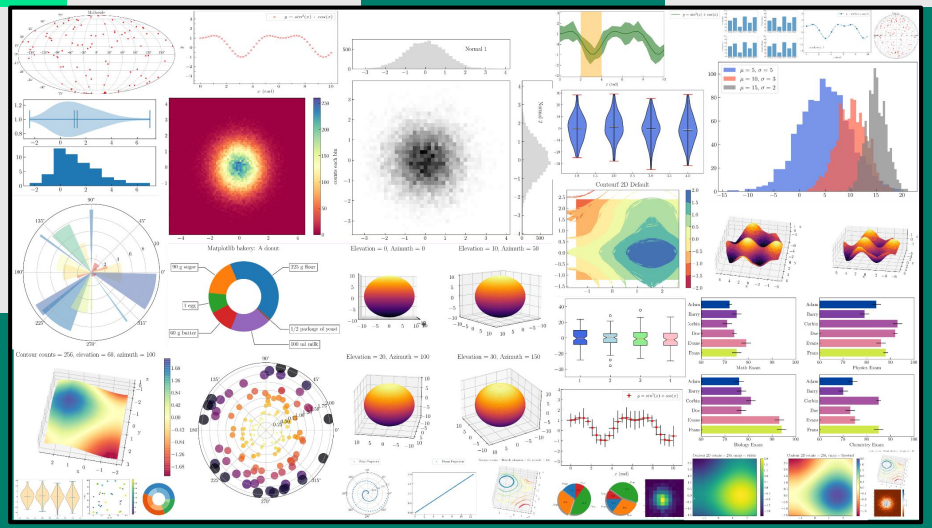
Matplotlib

IEEE EMBS 2026



A Python library that is like a free version of Matlab. This library can help you make a wide array of informative and beautiful graphs.

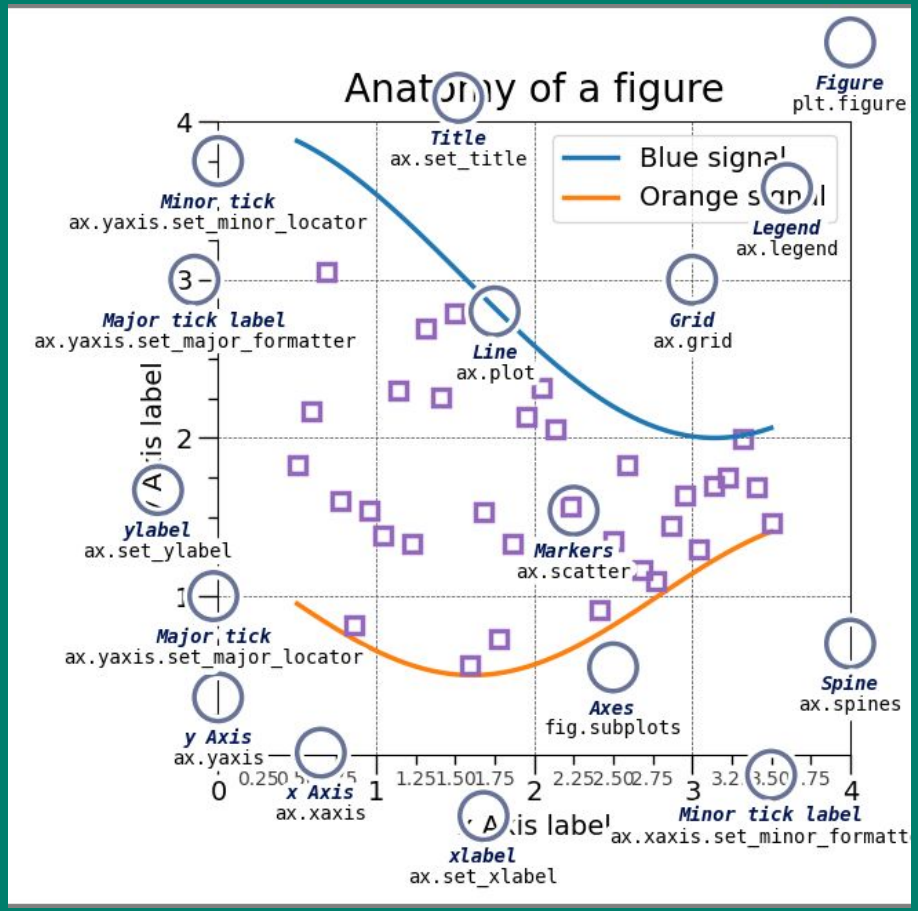


What *is* Matplotlib?

https://towardsdatascience.com/visualizations-with-matplotlib-part-1-c9651008b6b8/

What info should your graph have?

- TITLE <- what readers see first
- Axis labels (with units!!)
- Numbered tick marks
- Caption
  - Not added with matplotlib
- Legend (if 2+ categories)
- Depending on graph type:
  trendline, errors and error bars, hues, marker shapes, sizes, etc.



https://matplotlib.org/stable/gallery/showcase/anatomy.html
(Includes the source code, which is meta and cool!)

WHAT WE'RE ABOUT TO DO

1. Create your first graph in matplotlib to get the hang of things
2. Explore different methods for customizing your graph
3. Try different graph types and consider which ones may be most valuable



Part 1: Making your First Graph

(Before we start, make sure that you have the data to the right in an Excel spreadsheet)

Time (ms)	Distance (um)
0	5
2	1
4	-2
5	-1
8	2
10	3
12	2
13	4
16	6
17	1
20	2



Step 1	Step 2	Step 3	Step 4	Step 5
Create Python file and import libraries	Read Excel or CSV (comma-separated value) file	Set axes using the data	Set labels (title, axes, units, legend, etc.)	SHOW THE GRAPH with plt.show()

Can you ORDER the code on the next slide?

```
#read data
data_file = "C:\\Users\sonny\OneDrive\Desktop\Sonny College\\UCF\EMBS (IEEE) Workshop Chair (26-27)\\#3 - In
df = pd.read_excel(data_file) #set up DataFrame, which is how data transformed into graph
```

```
plt.xlabel("Time (ms)", fontname = "Times New Roman", fontsize = 14)
plt.ylabel("Distance (um)", fontname = "Times New Roman", fontsize = 14)
plt.title("Distance of Protein Motion", fontname = "Times New Roman", fontsize = 18)
```

```
#present at the end
#plt.legend() (not needed here)
plt.show()
```

```
#declare libraries to use
import matplotlib.pyplot as plt
import pandas as pd
```

BONUS: where would this block fit?

```
#color your graph (preferably disability-friendly)
plt.style.use("seaborn-v0_8-notebook") #colorblind-friendly style
```

```
#set axes and labels
x = df['Time (ms)']
y = df['Distance (um)']
plt.plot(x,y) #NEED a command like this to get things on axes
plt.axline((0,0), (20,0), color='k') #draws black x-axis
```

#2

```
#read data
data_file = "C:\\Users\sonny\OneDrive\Desktop\Sonny College\UCF\EMBS (IEEE) Workshop Chair (26-27)\\#3 - In
df = pd.read_excel(data_file) #set up DataFrame, which is how data transformed into graph
```

#4

```
plt.xlabel("Time (ms)", fontname = "Times New Roman", fontsize = 14)
plt.ylabel("Distance (um)", fontname = "Times New Roman", fontsize = 14)
plt.title("Distance of Protein Motion", fontname = "Times New Roman", fontsize = 18)
```

#5

```
#present at the end
#plt.legend() (not needed here)
plt.show()
```

#1

```
#declare libraries to use
import matplotlib.pyplot as plt
import pandas as pd
```

BONUS: where would this block fit?

ANSWER!

#4.5

```
#color your graph (preferably disability-friendly)
plt.style.use("seaborn-v0_8-notebook") #colorblind-friendly style
```

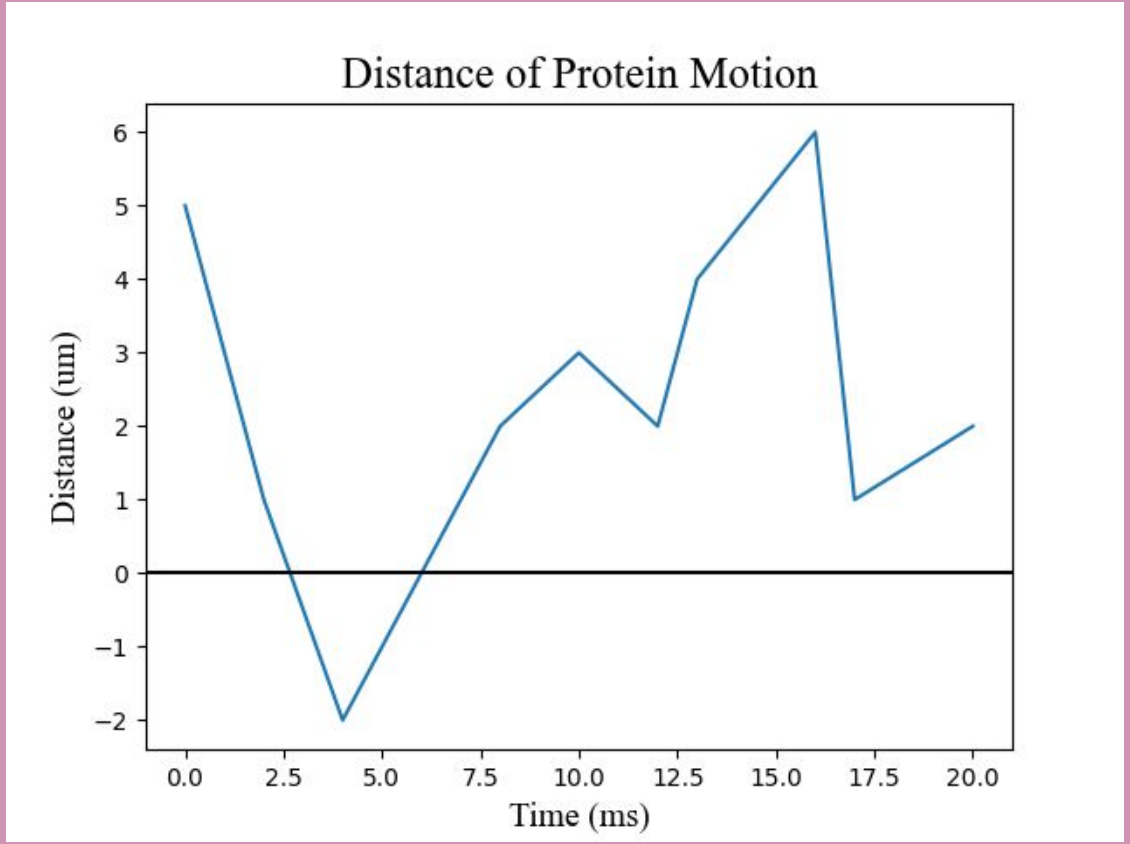
#3

```
#set axes and labels
x = df['Time (ms)']
y = df['Distance (um)']
plt.plot(x,y) #NEED a command like this to get things on axes
plt.axline((0,0), (20,0), color='k') #draws black x-axis
```

Now, let's see what the code makes!

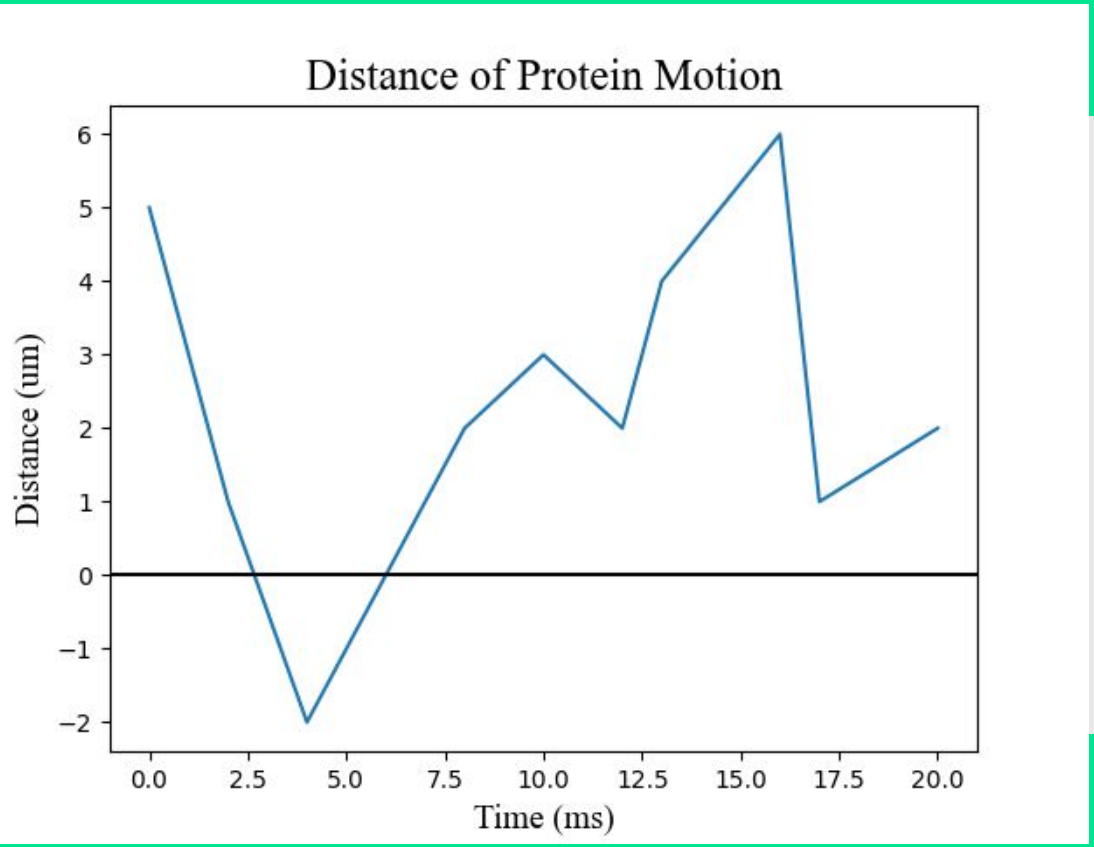


I will be going around the room to help answer any questions.



What if we try a different method?

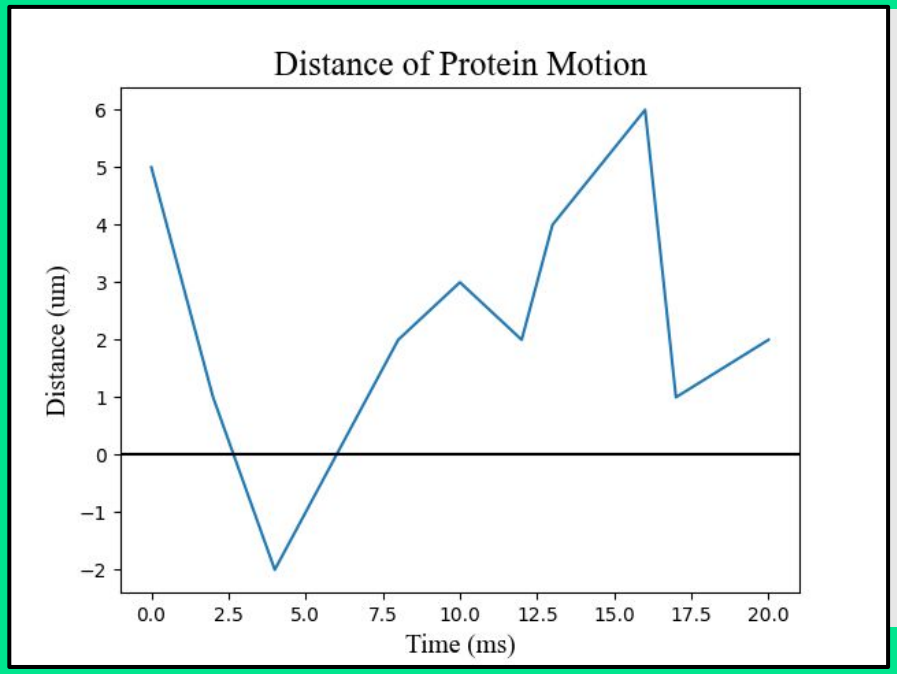
Part 2: Trying Different Methods



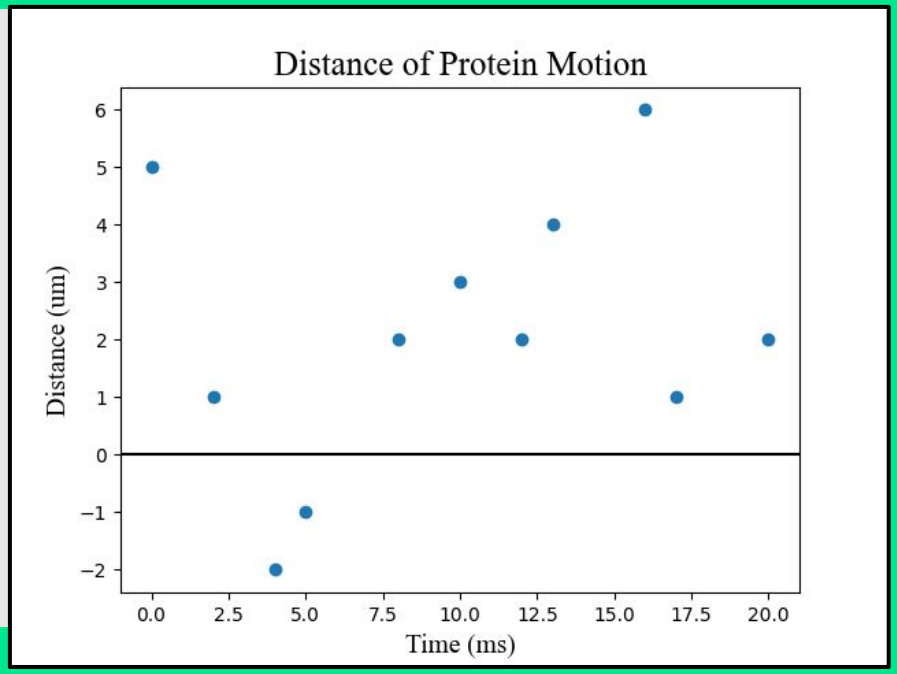
Replace “plot” with “scatter” in the code below.

```
#set axes and labels
x = df['Time (ms)']
y = df['Distance (um)']
plt.plot(x,y) #NEED a command
plt.axline((0,0), (20,0), color
```

plt.plot(x,y)



plt.scatter(x,y)



What do YOU want to change next?

Hue?

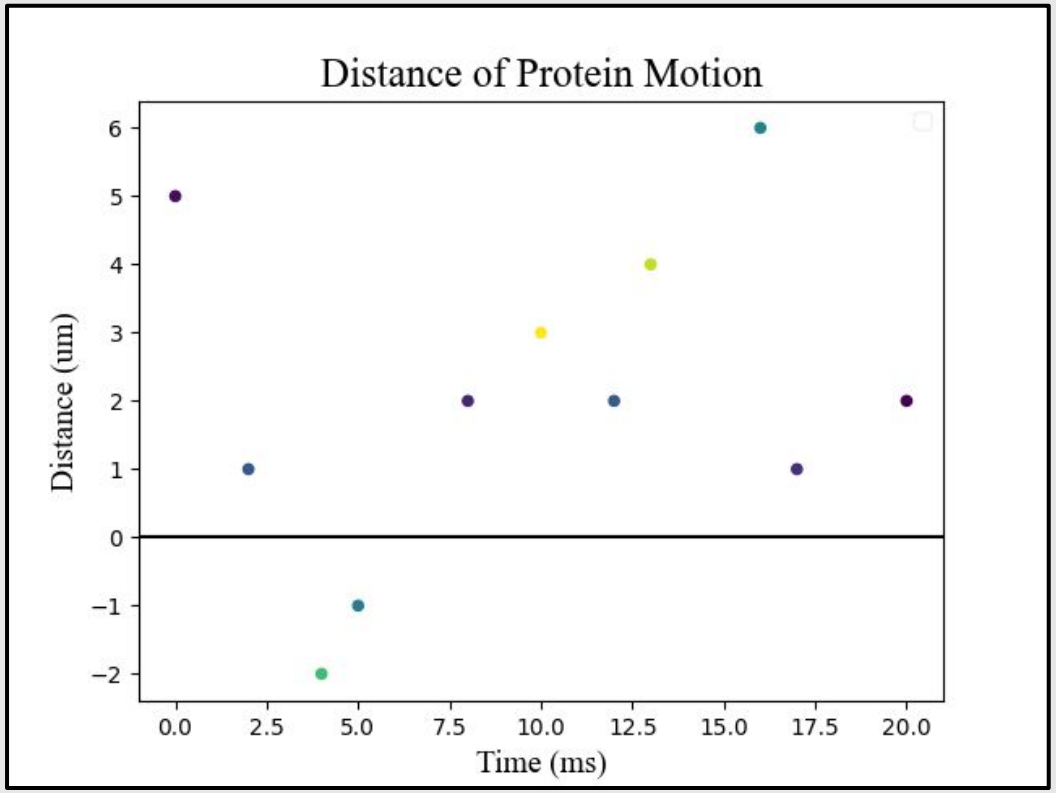
Background?

Marker Shape?

Grid?

Hue

← needs a third variable



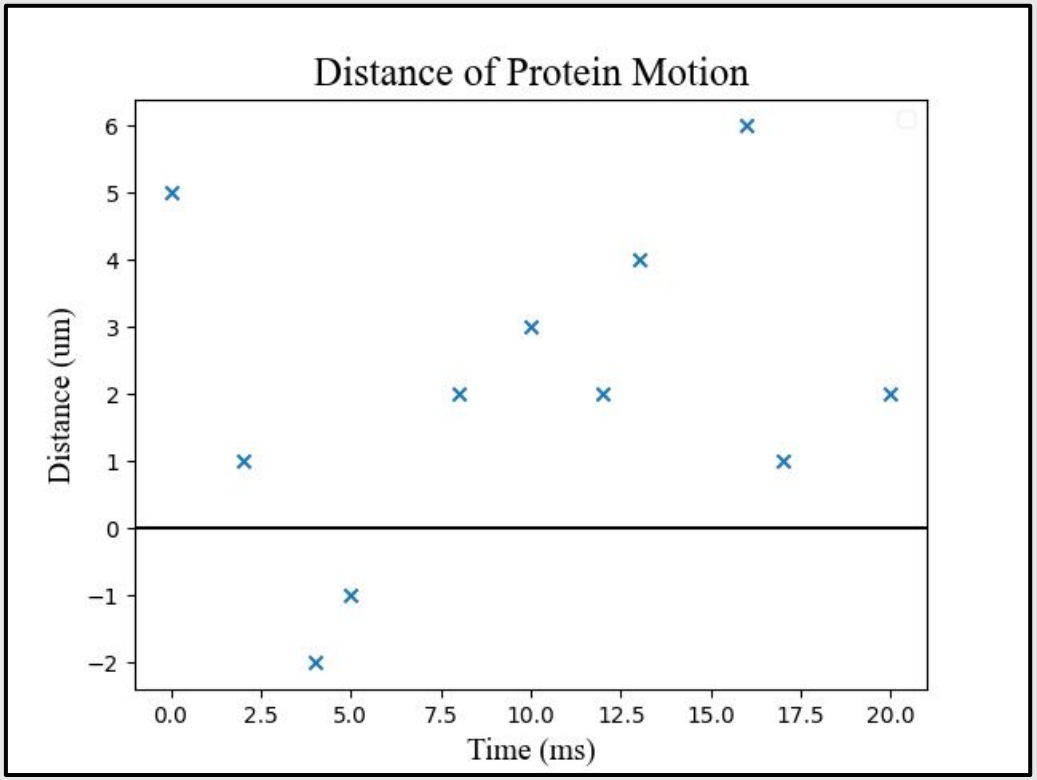
Add the ‘c’ parameter to the “scatter” method

```
c = df['Hydrodynamic Radius (nm)']
```

```
plt.scatter(x,y,c=c,s=20) #c = color, s = size (hue)
```

Bonus: What’s missing in this graph?

Marker Shape



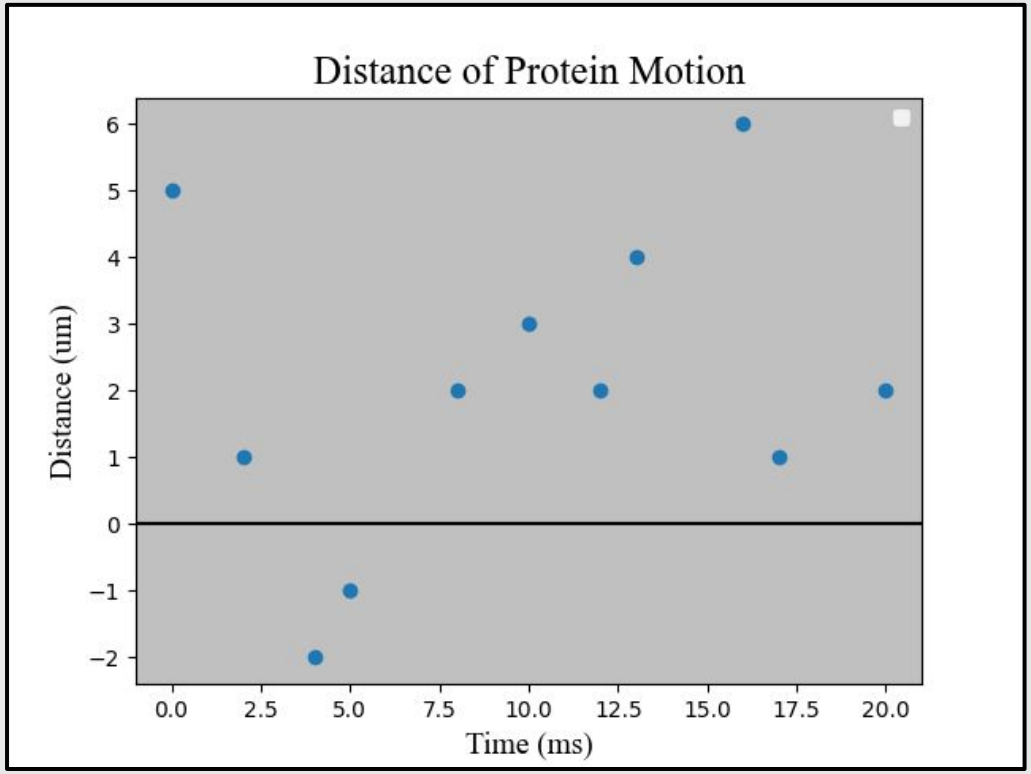
Add the ‘marker’ parameter to the “scatter” method

```
plt.scatter(x,y,marker='x') #(marker shape)
```

A few other marker types:

- ‘o’ for O
- ‘x’ for X
- ‘*’ for star
- ‘s’ for square
- ‘D’ for diamond
- etc.

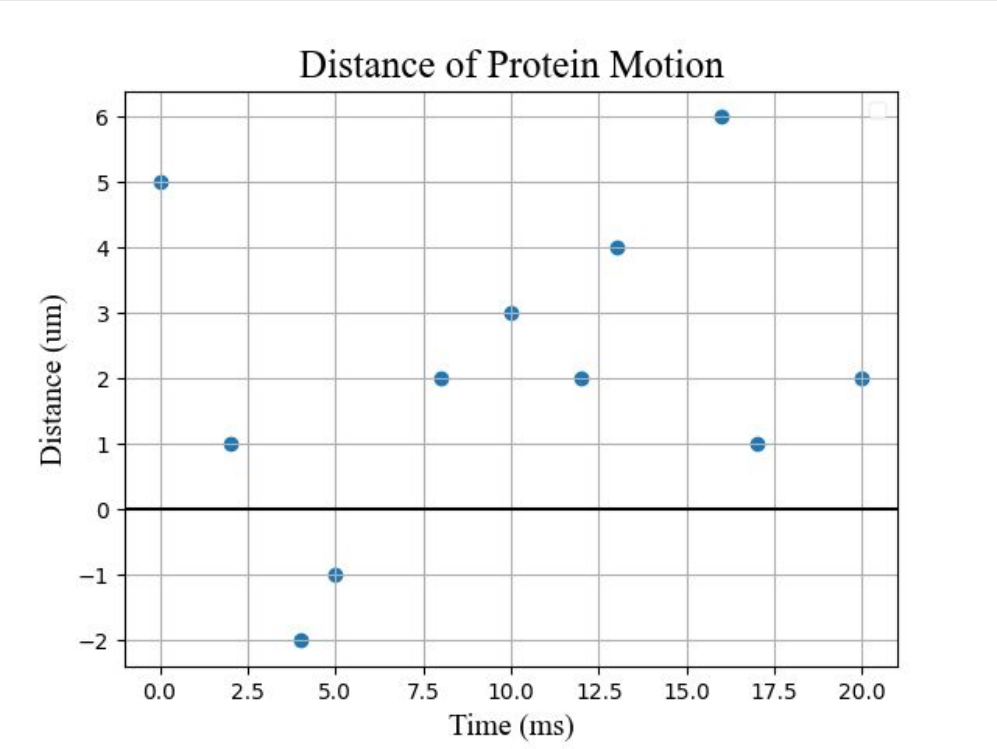
Background



Get the axes and set the face color BEFORE calling the “scatter” method

```
c = df['Hydrodynamic Radius (nm)']
ax = plt.axes() #(background) set BEFOR
ax.set_facecolor('silver')
plt.scatter(x,y) #what if you changed t
```

Grid



Just call plt.grid()

(*I personally prefer to do this by the code that sets the style*)

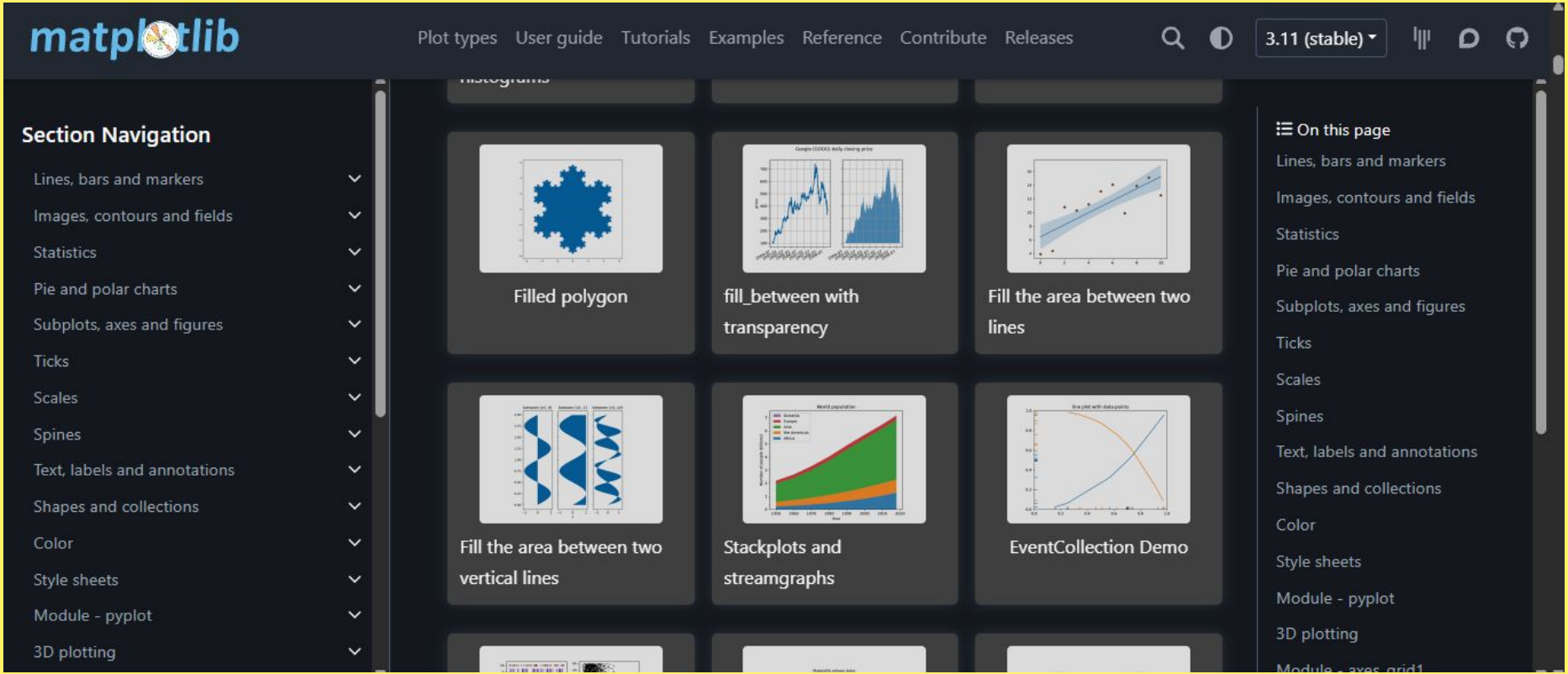
(*also this command can go in any location in the code*)

```
plt.style.us
plt.grid()
```

What other kinds of graphs can matplotlib make?

Part 3: Exploring Graph Types

https://matplotlib.org/stable/gallery/index.html



Choosing the Most Relevant Type of Graph For Your Data While Also Making It Look Nice and Impressive

- **Concise and precise communication?**
- Quantitative or Categorical data?
- Discrete or Continuous data?
  - Time Series - time is the x-variable; continuous
- Third Variable?
- Legend needed?

These are some of the main factors to consider, but other ones may come up



https://www.guvi.in/blog/fundamentals-of-matplotlib/

How to explore types of plots and their parameters: https://matplotlib.org/stable/plot_types/index.html

matplotlib Plot types User guide Tutorials Examples Reference Contribute Releases 3.11 (stable)

Section Navigation

- Pairwise data
- Statistical distributions
- Gridded data
- Irregularly gridded data
- 3D and volumetric data

Plot types

Plot types

Overview of many common plotting commands provided by Matplotlib.

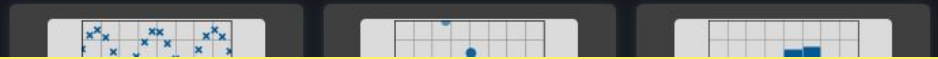
See the gallery for more examples and the tutorials page for longer examples.

On this page

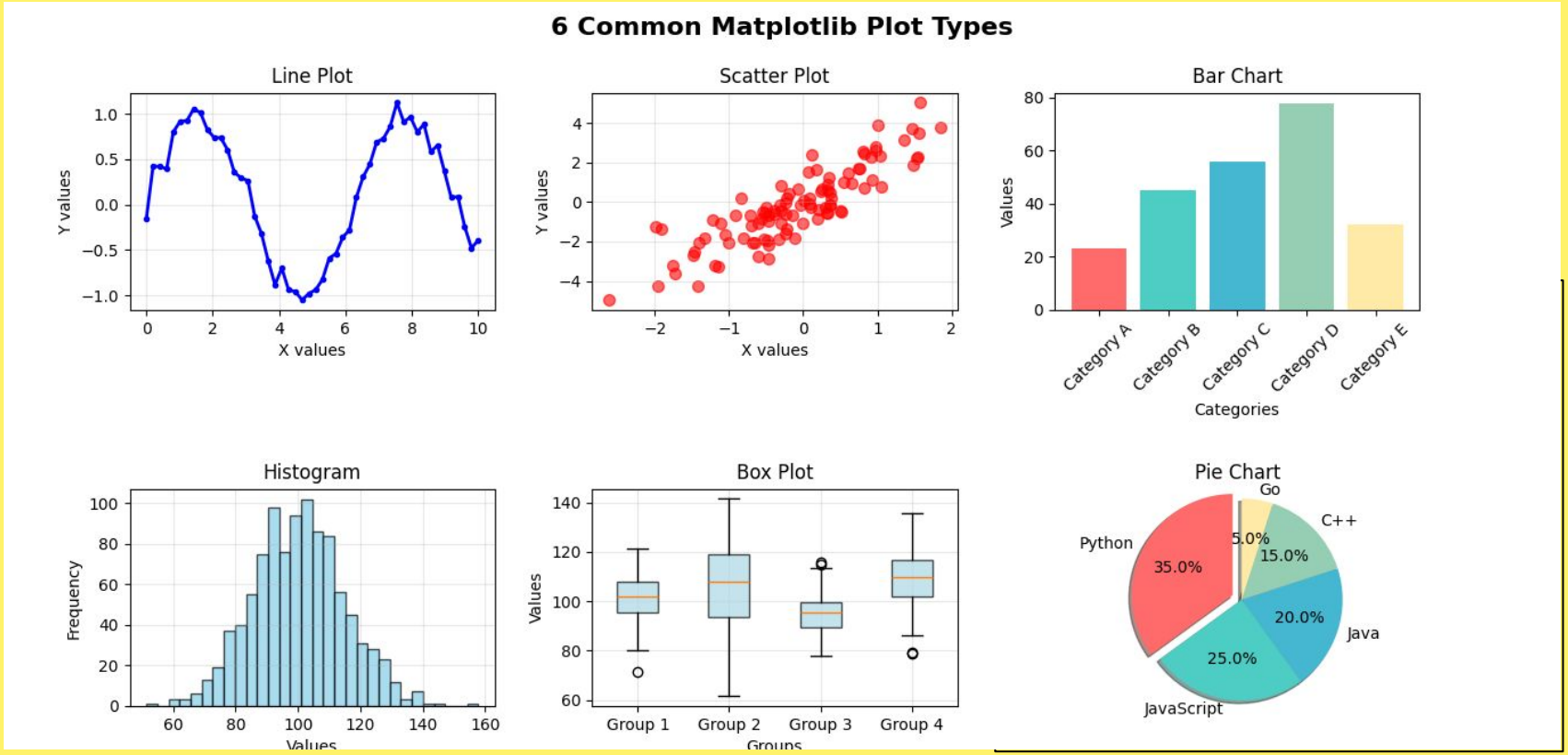
- Pairwise data
- Statistical distributions
- Gridded data
- Irregularly gridded data
- 3D and volumetric data

Pairwise data

Plots of pairwise (x, y), tabular $(var_0, \cdots, var_n)$, and functional $f(x) = y$ data.



Brief Overview of Key Graphs



https://www.pythonsnacks.com/p/matplotlib-6-common-plots

THANK YOU!